

Gregory Bian

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Computer Engineering (Co-op)

Expected May 2029

Vancouver, BC

Relevant Courses: Circuit Analysis, Computing Systems, Software Construction, Data Structures & Algorithms

TECHNICAL SKILLS

Design & Validation: SystemVerilog, FPGA, Siemens Questa, RISC-V Assembly, ARM Assembly

Hardware & Prototyping: Circuit Analysis, Oscilloscope, Soldering, CAN, SPI, UART

Programming & Frameworks: C, C++, Python, Java, JavaScript (React)

ENGINEERING STUDENT DESIGN TEAMS

Firmware Developer

Sep 2025 – Present

UBC Solar Racing

Vancouver, BC

- Collaborating with a cross-disciplinary team of 80+ software, electrical, and mechanical engineers, writing firmware in C for a solar-powered car, Cascadia, in preparation for FSGP and ASC solar racing competitions.
- Coordinated with the BMS and Power & Signals subteams to calibrate the accelerator pedal across a CAN Bus to the motor, implementing fail-safes, and **ensuring precise and reliable control in a safety-critical system**.
- Led the design and implementation of a multi-page LCD Screen over a STM32 microcontroller, consulting previous drivers for layout feedback and **improving visibility of real-time driving and diagnostic data by 30%**.
- Implemented memory-robust, FreeRTOS-based task scheduling on the LCD screen to **reduce latency by 20%** and enable concurrent operation of LCD updates with driver lights and turn signal updates.

Software Developer

Jan 2025 – Mar 2025

UVIC Aero

Victoria, BC

- Configured a Linux-based flight simulator to test autonomous flight functions for Firefly, a fire-extinguishing drone; documented WSL and terminal commands to **enable knowledge transfer to future team members**.
- Engineered a Python-based hotspot detection algorithm for infrared images using thresholding and connected components analysis in OpenCV, identifying over 85% of high-intensity regions in test datasets of 30+ images.

WORK EXPERIENCE

STEM and Computing/Robotics Instructor

Jun 2025 – Aug 2025

UBC Geering Up Outreach

Vancouver, BC

- Led STEM-based Summer Camps for Geering Up, a STEM Outreach Program at the Faculty of Applied Science at UBC, **reaching over 28000+ youth a year**.
- Instructed up to 30 students per week on Arduino-based embedded programming, computer architecture, and data analysis in Python, explaining complex technical concepts in a fun and engaging way for youth.

PROJECTS

Single-Cycle RISC-V CPU | *SystemVerilog, FPGA, RISC-V Assembly, Questa*

Nov 2025 – Dec 2025

- Architected a single-cycle CPU in SystemVerilog, executing RV32 machine code instructions to perform arithmetic and drive 7-segment displays and LEDs from switch inputs on a DE10-Lite FPGA.
- Developed a SystemVerilog testbench to verify hardware functionality using waveform analysis in Siemens Questa and confirm datapath and control logic correctness during instruction execution.

Combination Lock and Keypad | *SystemVerilog, FPGA*

Nov 2025

- Designed and implemented a SystemVerilog finite state machine for a combination-lock module simulating a vault safe, supporting password entry, cancellation, and lock/unlock states with real-time 7-segment display updates.
- Synchronized clock signals between two DE10-Lite FPGA boards, enabling reliable parsing of keypress data from the external keyboard scanning module.

Bare-Metal Tron Game | *C, RISC-V Assembly, FPGA*

Nov 2025

- Developed a bare-metal Tron game in C, generating VGA output for real-time graphics rendering on a DE10-Lite FPGA and featuring multiplayer or single-player against a simple logic-driven robot.
- Implemented hardware interrupt-driven control logic using RISC-V Assembly to handle player and robot movement, collision detection (deaths), and real-time HEX display updates.